

To achieve an n -th (n even) order of accuracy in the first derivative of $f(x)$ at a point x using a non-uniform mesh, the strategy is use a $n + 1$ -point stencil. Usually, one of the points is x itself, and the remaining n points are $x + h_1$, $x + h_2$, etc., up to $x + h_n$. Each discretization h_i can be positive or negative. Typically for internal nodes, $n/2$ points are negative (to the left of x) and the order $n/2$ points are positive. Boundary or near-boundary nodes will require that more points be taken from the side opposite the boundary.

Then we need the Taylor-expansion up to order n at every point $x + h_i$:

$$f(x + h_i) = \sum_{k=0}^n \frac{h_i^k}{k!} f^{(k)}(x) + \mathcal{O}(h^{n+1}), \quad (1)$$

assuming $h_0 = 0$ and $h_0^0 = 1$.

We can approximate the first derivative as a linear combination of each $f(x + h_i)$:

$$f'(x) = \sum_{i=0}^n A_i f(x + h_i), \quad (2)$$

and substituting in the Taylor expansion for each $f(x + h_i)$:

$$f'(x) = \sum_{k=0}^n \sum_{i=0}^n \frac{h_i^k}{k!} A_i f^{(k)}(x) + \mathcal{O}(h^n). \quad (3)$$

The order of error has been reduced by 1 since we expect each A_i to be on the order of $\mathcal{O}(\frac{1}{h_i})$. This is equal to the order of accuracy of the finite difference approximation.

If we expand Equation 3 to a few orders of derivative:

$$\begin{aligned} f'(x) = & \left[\sum_{i=0}^n A_i \right] f(x) + \left[\sum_{i=0}^n h_i A_i \right] f'(x) \\ & + \left[\frac{1}{2} \sum_{i=0}^n h_i^2 A_i \right] f''(x) + \left[\frac{1}{6} \sum_{i=0}^n h_i^3 A_i \right] f'''(x) + \dots \end{aligned} \quad (4)$$

In order to equate left-hand side of Equation 4 with the right-hand side, every single bracket must be zero except for the one in front of $f'(x)$, which must be 1. We can solve for the unknown A_i 's using a system of equations:

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & h_1 & h_2 & \dots & h_n \\ 0 & h_1^2 & h_2^2 & \dots & h_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & h_1^n & h_2^n & \dots & h_n^n \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (5)$$

The matrix in Equation 7 is ill-conditioned for general choices of h_i . But for derivative approximation, the h 's are sufficiently well-spaced so that $\frac{\max |h_i|}{\min |h_i|}$

is within one order-of-magnitude. Every row k of this matrix can be scaled by $\frac{1}{h_r^k}$, where h_r is the mean $|h_i|$, so that the conditioning can be improved.

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & h_1/h_r & h_2/h_r & \cdots & h_n/h_r \\ 0 & (h_1/h_r)^2 & (h_2/h_r)^2 & \cdots & (h_n/h_r)^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & (h_1/h_r)^n & (h_2/h_r)^n & \cdots & (h_n/h_r)^n \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix} = \frac{1}{h_r} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (6)$$

As noted in Equation 3, the order of accuracy in estimating the first derivative is one less than the number of stencils used. If the same Taylor expansion of order n is used to estimate the second derivative, two orders of accuracy is lost, leading to an only an order of accuracy of only $n - 1$. Using a symmetric stencil around the point of interest will bring the order of accuracy back up to n , but for a general, asymmetric and non-uniform mesh, a Taylor expansion of order $n + 1$ is required.

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & h_1/h_r & h_2/h_r & \cdots & h_{n+1}/h_r \\ 0 & (h_1/h_r)^2 & (h_2/h_r)^2 & \cdots & (h_{n+1}/h_r)^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & (h_1/h_r)^n & (h_2/h_r)^n & \cdots & (h_{n+1}/h_r)^{n+1} \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \\ \vdots \\ A_{n+1} \end{bmatrix} = \frac{2}{h_r^2} \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad (7)$$

For uniformly-spaced grids with step size h between adjacent points, Tables 1 and 2 lists the coefficients used to approximate the first- and second-derivatives, respectively, up to 6 orders of accuracy. The column *L-R* denotes how many points are to the left and right of the point where the derivative is taken. First-derivative formulae are anti-symmetric, so the coefficients of a *R-L* scheme are negative the corresponding *L-R* coefficients. Second-derivative formulae are symmetric, so the *R-L* coefficients are identical the corresponding *L-R* counterparts.

Table 1: Coefficients and step sizes for a uniformly-spaced first-derivative approximation

n	$L-R$		i						
			0	1	2	3	4	5	6
2	1-1	h_i/h	0	-1	1				
		$A_i \cdot 2h$	0	-1	1				
2	0-2	h_i/h	0	1	2				
		$A_i \cdot 2h$	-3	4	-1				
4	2-2	h_i/h	0	-2	-1	1	2		
		$A_i \cdot 12h$	0	1	-8	8	-1		
4	1-3	h_i/h	0	-1	1	2	3		
		$A_i \cdot 12h$	-10	-3	18	-6	1		
4	0-4	h_i/h	0	1	2	3	4		
		$A_i \cdot 12h$	-25	48	-36	16	-3		
6	3-3	h_i/h	0	-3	-2	-1	1	2	3
		$A_i \cdot 60h$	0	-1	9	-45	45	-9	1
6	2-4	h_i/h	0	-2	-1	1	2	3	4
		$A_i \cdot 60h$	-35	2	-24	80	-30	8	-1
6	1-5	h_i/h	0	-1	1	2	3	4	5
		$A_i \cdot 60h$	-77	-10	150	-100	50	-15	2
6	0-6	h_i/h	0	1	2	3	4	5	6
		$A_i \cdot 60h$	-147	360	-450	400	-225	72	-10

Table 2: Coefficients and step sizes for a uniformly-spaced second-derivative approximation

n	$L-R$		i							
			0	1	2	3	4	5	6	7
2	1-1	h_i/h	0	-1	1					
		$A_i \cdot h^2$	-2	1	1					
2	0-3	h_i/h	0	1	2	3				
		$A_i \cdot h^2$	2	-5	4	-1				
4	2-2	h_i/h	0	-2	-1	1	2			
		$A_i \cdot 12h^2$	-30	-1	16	16	-1			
4	1-4	h_i/h	0	-1	1	2	3	4		
		$A_i \cdot 12h^2$	-15	10	-4	-14	-6	1		
4	0-5	h_i/h	0	1	2	3	4	5		
		$A_i \cdot 12h^2$	45	-154	214	-156	61	-10		
6	3-3	h_i/h	0	-3	-2	-1	1	2	3	
		$A_i \cdot 180h^2$	-490	2	-27	270	270	-27	2	
6	2-5	h_i/h	0	-2	-1	1	2	3	4	5
		$A_i \cdot 180h^2$	-378	-11	214	130	85	-54	16	-2
6	1-6	h_i/h	0	-1	1	2	3	4	5	6
		$A_i \cdot 180h^2$	-70	126	-486	855	-670	324	-90	11
6	0-7	h_i/h	0	1	2	3	4	5	6	7
		$A_i \cdot 180h^2$	938	-4014	7911	-9490	7380	-3618	1019	-126